

AAYUSH RATHOD

Senior Java Backend Developer | Microservices & Distributed Systems | Banking & FinTech

Toronto, ON • +1 (416) 731-2864 • aayushrathod521@gmail.com

linkedin.com/in/aayush-rathod11 • github.com/aayush521 • aayush521.github.io

Authorized to work in Canada — Open Work Permit (valid through Aug 2027)

PROFESSIONAL SUMMARY

Senior Java Developer with 8+ years of experience designing and delivering enterprise-grade microservices, REST and SOAP web services, and distributed systems for global banking and financial institutions. Proven record of architecting AML/ATF, Anti-Fraud, Core Banking, and IFRS 9 platforms across 7+ multi-country implementations, with regulatory compliance frameworks aligned to FATF, OFAC, UN, EU and analogous to Canadian OSFI/FINTRAC/PCMLTFA standards. Deep expertise in Java, Spring Boot, Microservices Design Patterns, Service Oriented Architecture, Oracle, MongoDB, Docker, and Kubernetes. Targeting Senior Backend / Java / Microservices roles in Canadian FinTech and enterprise software organizations.

TECHNICAL SKILLS

Programming & Backend: Java (8/11/17/21), Spring Boot, Spring MVC, Spring WebFlux, Spring Data JPA, Spring Security (JWT/OAuth2), Java Standard Library, REST Web Services, SOAP Web Services, Microservices Architecture, OOP, Data Structures, Algorithms, Python, Shell Scripting

Architecture & Design: Microservices Design Patterns (API Gateway, Circuit Breaker, Saga, CQRS, Service Discovery), Distributed Systems Architecture, Service Oriented Architecture, Enterprise Application Design, System Integration, Event-Driven Architecture, Software Architecture Planning, Object-Oriented Design Principles

Databases: Oracle 11g/12c/19c (PL/SQL, Query Tuning, Partitioning), MongoDB / MongoDB Atlas, SQL Server, SQL, Database Design, Data Persistence, Aggregation Pipelines

Cloud & Containerization: AWS (EC2, S3, Lambda, RDS, IAM, CloudWatch), Azure, Docker, Kubernetes, Containerized Microservices Deployment, Apache Tomcat, JBoss, Cloudflare (R2, Stream)

DevOps & Tools: TeamCity, Jenkins, GitHub Actions, Git, SVN, JIRA, Confluence, Build Automation, Version Control, CI/CD Pipelines, Maven, Gradle, Swagger/OpenAPI

Messaging & Integration: Apache Kafka, REST APIs, SOAP, Pentaho PDI, Informatica ETL, Webhooks, Stripe Payment Gateway, OAuth 2.0

Practices & Banking Domain: Agile Software Development, Scrum Methodology, Secure Coding Practices, Code Reviews, TDD, Stakeholder Collaboration, L2/L3 Production Support; AML/ATF, Anti-Fraud (AFS), Core Banking (CBS), IFRS 9 / ECL, KYC, Sanctions & Watchlist Screening (UN/OFAC/EU/HMT), SWIFT / NEFT / RTGS, Regulatory Compliance (FATF, OSFI, FINTRAC, PCMLTFA equivalents)

KEY ACHIEVEMENTS

- **Multi-Country Delivery:** Delivered 7+ enterprise banking implementations across Oman, Saudi Arabia, Dubai, Philippines, Mauritius, Bhutan, and Nepal — owning UAT, multi-country go-lives, and post-production stabilization.
- **Performance Engineering:** Reduced ETL batch run time by 40% through PL/SQL query optimization and pipeline redesign, directly improving SLA compliance.
- **Fraud Detection at Scale:** Designed MongoDB aggregation pipelines that identified 90%+ of fraudulent transactions in a live Anti-Fraud System.
- **Event-Driven Architecture:** Architected Apache Kafka-based event streaming for real-time fraud alerts and transaction monitoring across distributed banking microservices.

PROFESSIONAL EXPERIENCE

Senior Backend Engineer (Freelance)

Nov 2024 – Present

PrecastXchange — B2B Marketplace Platform (independent freelance engagement) | Toronto, ON

- Designed and built backend microservices using Java 21 and Spring Boot for a B2B marketplace platform, applying microservices design patterns, clean OOP, and a service-oriented structure.
- Implemented OAuth 2.0 (Google) authentication, JWT-secured REST APIs, and role-based access control across all endpoints, applying secure coding practices and OWASP Top 10 mitigations.
- Integrated Stripe payment gateway end-to-end (checkout, webhooks, idempotent transaction handling, refund flows) with zero duplicate-charge incidents in production.
- Designed MongoDB Atlas schemas, compound indexes, and aggregation pipelines optimized for high-read marketplace workloads, reducing query latency by ~60% on listing endpoints.

- Containerized services with Docker and built video upload/streaming workflows on Cloudflare R2 and Cloudflare Stream, enabling secure on-demand media delivery.
- Set up CI/CD pipelines with GitHub Actions for automated testing, build, and deployment, shortening release cycles from days to hours.

Java Backend Developer / Associate Consultant

Nov 2021 – Oct 2024

Infrasoft Technologies Ltd (FinTech vendor to global tier-1 banks) | Mumbai, India

- Designed and deployed microservices for AML/ATF and Anti-Fraud (AFS) platforms using Java, Spring Boot, Spring MVC, REST and SOAP web services, and Spring Data JPA across distributed multi-country banking environments — compliance frameworks aligned to FATF, OFAC, UN, EU, and analogous to OSFI / FINTRAC / PCMLTFA standards.
- Applied microservices design patterns (API Gateway, Circuit Breaker, Service Discovery, Saga) and SOA principles to integrate Core Banking (CBS), AML, and reporting modules across enterprise estates.
- Built customer onboarding and KYC microservices with real-time UN/OFAC/EU/HMT sanctions and watchlist screening, ensuring regulatory compliance for global banking clients.
- Integrated Apache Kafka for transaction streaming, event-driven fraud alerts, and asynchronous inter-service communication across AML/AFS modules.
- Engineered ETL pipelines with Pentaho PDI and Informatica to feed Core Banking and third-party data into regulatory risk and credit-loss analytics.
- Authored PL/SQL procedures and packages for IFRS 9, automating Expected Credit Loss (ECL) computation and provisioning across all three financial stages.
- Tuned MongoDB clusters and aggregation queries for fraud analytics and high-volume transaction monitoring (90%+ fraud identification rate).
- Established CI/CD pipelines in Jenkins and GitHub Actions; introduced unit and integration test suites that improved deployment reliability and reduced release defects.
- Delivered L3 production support — RCA, log analysis, DB tuning, and UAT/production deployment validation; collaborated with global stakeholders to deliver multi-country go-lives.

Software Developer (Java + PL/SQL + COBOL) / L3 Support

Feb 2020 – Oct 2021

Onward e-Services Ltd (Client: Bank of Maharashtra — ~30M+ customers, 2,200+ branches) | Pune, India

- Developed Java and PL/SQL backend modules for Remittances, Payments, and Core Banking, including high-volume SWIFT, NEFT, RTGS, and ATM file-based transaction processing.
- Maintained EOD/SOD batch processes and provided L3 support — log analysis, RCA, and DevOps-coordinated deployments — to meet stringent banking SLAs and regulatory cut-offs.

Assistant Systems Analyst (Java + PL/SQL + COBOL)

Mar 2018 – Feb 2020

C-Edge Technologies Ltd (TCS-SBI JV; Client: NABARD-supervised cooperative banks) | Mumbai, India

- Built and supported General Ledger, Deposits, and Trickle Feed modules using PL/SQL, COBOL, and Java for cooperative banking infrastructure serving 100+ regional banks.
- Authored Python and Shell scripts for Autosys batch monitoring; delivered SQL performance improvements and on-site post-implementation support.

EDUCATION

B.E., Computer Engineering — Pune Vidyarthi Griha's College of Engineering, Pune, India

June 2017

Diploma in Computer Technology — K. K. Wagh College of Polytechnic, Nashik, India

June 2014

Certifications: AWS Certified Cloud Practitioner — Amazon Web Services